

Reg. No. :

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Question Paper Code : 11328

B.E./B.Tech. DEGREE EXAMINATION, NOVEMBER/DECEMBER 2012.

Fourth Semester

Electronics and Communication Engineering

EC 2253/147403/EC 43/10144/EC 404/EC 1253/080290021 - ELECTROMAGNETIC FIELDS

(Regulation 2008)

Time : Three hours

Maximum : 100 marks

Answer ALL questions.

PART A — (10 × 2 = 20 marks)

1. State Divergence theorem.
2. What is the significance of electric flux density?
3. Write Lorentz's force equation for a moving charge.
4. Define the term 'Magnetic flux density'.
5. What do you understand from current continuity equation?
6. Define the term 'Relative permeability'.
7. Define Faraday's law of Electromagnetic induction.
8. An EM wave has E_x and H_y as components of electric and magnetic fields respectively. Find the direction of power flow.
9. What do you mean by a wave?
10. An EM wave has electric component given by, $E = E_0 \sin(\alpha x - \beta z)(\vec{a}_x + \vec{a}_z)$ V/m. Comment on the polarization of the wave.

PART B — (5 × 16 = 80 marks)

11. (a) (i) Find the electric field intensity at a point P located at (0,0,h) m due to charge of surface charge density σ C/m² uniformly distributed over the circular disc $r \leq a, z = 0$ m. (10)
- (ii) Determine the divergence and curl of the given field $F = 30\vec{a}_x + 2xy\vec{a}_y + 5xz^2\vec{a}_z$ at (1,1,-0.2) and hence state the nature of the field. (6)

Or

- (b) (i) Point charges Q and $-Q$ are located at $(0,0,d/2)$ and $(0,0,-d/2)$. Show that the potential at a point (r,θ,ϕ) is inversely proportional to r^2 noting that $r \gg d$. (8)
- (ii) Given a field $E = \frac{-6y}{x^2}a_x + \frac{6}{x}a_y + 5a_z$ V/m, find the potential difference V_{AB} between $A(-7,2,1)$ and $B(4,1,2)$. (8)
12. (a) (i) Derive an expression for magnetic field intensity due to a linear conductor of infinite length carrying current I at a distant point P . Assume R to be the distance between conductor and point P . Use Biot Savart's law. (8)
- (ii) Derive an expression for magnetic field intensity on the axis of a circular loop of radius ' a ' carrying current I . (8)
- Or
- (b) (i) Obtain the expressions for scalar and vector magnetic potential. (8)
- (ii) At a point $P(x,y,z)$ the components of vector magnetic potential $\vec{A}$ are given as $A_x = 4x + 3y + 2z$, $A_y = 5x + 6y + 3z$ and $A_z = 2x + 3y + 5z$. Determine the magnetic flux density $\vec{B}$ at the point P . (4)
- (iii) Given the magnetic flux density $\vec{B} = 2.5 \left(\sin \frac{\pi x}{2} \right) e^{-2y} \vec{a}_z$ Wb/m², find the total magnetic flux crossing the strip defined by $z = 0$, $y \geq 0$, $0 \leq x \leq 2m$. (4)
13. (a) (i) Determine whether or not the following potential fields satisfy the Laplace's equation.
- (1) $V = x^2 - y^2 + z^2$ (2)
- (2) $V = r \cos \phi + z$ (3)
- (3) $V = r \cos \theta + \phi$. (3)
- (ii) Solve the Laplace's equation for the potential field in the homogenous region between the two concentric conducting spheres with radius ' a ' and ' b ' where $b > a$, $V = 0$ and $r = b$ and $V = V_0$ at $r = a$. Find the capacitance between the two concentric spheres. (8)

Or

- (b) (i) Derive the expression for the inductance of a toroidal coil with N turns, carrying current I and the radius of the toroid R. (8)
- (ii) Considering a toroidal coil, derive an expression for energy density. (8)
14. (a) (i) Electric flux density in a charge free region is given by $\vec{D} = 10x\hat{a}_x + 5y\hat{a}_y + kz^2\hat{a}_z \mu C/m^2$, find the constant k. (6)
- (ii) If electric field intensity in free space is given by $\vec{E} = \frac{50}{\rho} \cos(10^8t - 10z)\hat{a}_\rho V/m$. Find the magnetic field intensity $\vec{H}$. (10)

Or

- (b) (i) State and prove Poynting theorem. (8)
- (ii) Derive the expression for total power flow in co-axial cable. (8)
15. (a) (i) From the Maxwell's equation, derive the electromagnetic wave equation in conducting medium for E and H fields. (10)
- (ii) Calculate the attenuation constant and phase constant for the uniform plane wave with the frequency of 100 GHz in a conducting medium for which $\mu_r = 1$ and $\sigma = 58 \times 10^6 S/m$. (6)

Or

- (b) (i) With reference electro magnetic waves, explain the following:
- (1) linear polarization (3)
 - (2) circular polarization (3)
 - (3) elliptical polarization (2)
- (ii) A plane wave is incident, normally on a perfect conductor. Derive the expression for standing wave. Find the location of nodes and antinodes in E and H fields. Sketch the Standing wave pattern. (8)

Reg. No. : _____

Question Paper Code : 21358

B.E./B.Tech. DEGREE EXAMINATION, MAY/JUNE 2013.

Fourth Semester

Electronics and Communication Engineering

EC 2253/EC 43/10144 EC 404/EC 1253/080290021 — ELECTROMAGNETIC
FIELDS

(Regulation 2008/2010)

Time : Three hours

Maximum : 100 marks

Answer ALL questions.

PART A — (10 × 2 = 20 marks)

1. Define electric field and electric potential.
2. State divergence theorem.
3. State Biot-Savart's law.
4. Define magnetic vector potential.
5. Determine the capacitance of the parallel plate capacitor composed of tin foil sheets, 25cm square for plates separated through a glass dielectric 0.5cm thick with relative permittivity of 6.
6. State point form of Ohm's law.
7. Distinguish between conduction current and displacement current.
8. Write down the expressions for instantaneous and complex Poynting vector.
9. Find the skin depth at a frequency of 3MHz in aluminium where $\sigma = 38.2 \text{ M s/m}$ and $\mu_r = 1$.
10. What is Brewster angle?

PART B — (5 × 16 = 80 marks)

11. (a) Derive an expression for the electric field due to a straight and infinite uniformly charged wire of length 'L' meters and with a charge density of $+ \rho_c/m$ at a point P which lies along the perpendicular bisector of wire. (16)

Or

- (b) (i) A uniform line charge $\rho_L = 25 \text{ Nc/m}$ lies on the $x = 3\text{m}$ and $y = 4\text{m}$ in free space. Find the electric field intensity at a point (2, 3 and 15) m. (8)
- (ii) Given that potential $V = 10 \sin \theta \cos \Phi / r^3$ find the electric flux density D at $(2, \pi/2, 0)$. (8)
12. (a) (i) Derive an expression for force between two current carrying conductors. (8)
- (ii) An iron ring with a cross sectional area of 3 cm square and mean circumference of 15cm is wound with 250 turns wire carrying a current of 0.3A. The relative permeability of ring is 1500. Calculate the flux established in the ring. (8)

Or

- (b) Derive the expressions for magnetic field intensity and magnetic flux density due to finite and infinite line carrying a current I. (16)
13. (a) Derive the boundary conditions of the normal and tangential components of electric field at the interface of two media with different dielectrics. (16)

Or

- (b) The capacitance of the conductor formed by the two parallel metals sheets, each 100cm^2 , in area separated by a dielectric 2 mm thick is 2×10^{-10} micro farad. A potential of 20kv is applied to it. Find
- (i) Electric flux (4)
- (ii) Potential gradient in kV/m (4)
- (iii) The relative permittivity of materials (4)
- (iv) Electric flux density. (4)

14. (a) With necessary explanation, derive the Maxwell's equation in differential and integral forms. (16)

Or

- (b) (i) The conduction current flowing through a wire with conductivity $\sigma = 3 \times 10^7 \text{ S/m}$ and the relative permeability $\epsilon_r = 1$ is given by $I_c = 3 \sin \omega t \text{ (mA)}$. If $\omega = 10^8 \text{ rad/sec}$, find the displacement current. (8)
- (ii) An electric field in a medium which is source free is given by $E = 1.5 \cos(10^8 t - \beta z) \hat{a}_x \text{ V/m}$. Find B, H and D. Assume $\epsilon_r = 1, \mu_r = 1, \sigma = 0$. (8)
15. (a) A plane sinusoidal electromagnetic wave traveling in space has $E_{\max} = 150 \mu\text{V/m}$. (16)
- (i) Find the accompanying $H_{\max}$.
- (ii) Propagation is in X direction and H is oriented in Y direction. What is the direction of E?
- (iii) Compute the average power transmitted.

Or

- (b) Explain in detail on what happens when the wave is incident
- (i) Normally on perfect conductor (8)
- (ii) Obliquely to the surface of perfect dielectrics. (8)

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Question Paper Code : 51400

B.E./B.Tech DEGREE EXAMINATION, MAY/JUNE 2014

Fourth Semester

Electronics and Communication Engineering

EC 2253/EC 43/EC 1253/080290021/10144 EC 101 — ELECTROMAGNETIC
FIELDS

(Regulation 2008/2010)

Time : Three hours

Maximum : 100 marks

Answer ALL questions

PART A — (10 × 2 = 20 marks)

1. In XY plane, $Q_1 = 100 \mu\text{C}$ at (2, 3)m experiences a repulsive force of 7.5N because of Q_2 at (10, 6)m. Find Q_2 .
2. What is Gradient?
3. If the magnetic field $B = 25x\hat{i} - 12y\hat{j} - \alpha z\hat{k}$ (T), find α .
4. Write Biot-Savart law.
5. An infinite solenoid (n turns per unit length, current I) is filled with a linear material of susceptibility χ_m . Find the magnetic field inside the solenoid.
6. Write the boundary conditions for electric field.
7. Find the Poynting vector on the surface of a long straight conducting wire of radius 'b' and conductivity σ that carries a direct current I.
8. State the flux rule for a nonrectangular loop moving through a nonuniform magnetic field.
9. A sinusoidal electric intensity of amplitude 250 V/m and frequency 1 GHz exists in a lossy dielectric medium that has a relative permittivity of 2.5 and loss tangent of 0.001. Find the effective conductivity of the lossy medium.
10. What is skin depth?

PART B — (5 × 16 = 80 marks)

11. (a) (i) State and Explain the fundamental theorems of Divergence and Curl. (8)
 (ii) Find the electric field at a distance 'z' above the center of a flat circular disc of radius R, which carries a uniform surface charge σ . (8)

Or

- (b) (i) Get the relationship between potential and electric field. A (physical) dipole consists of two equal and opposite charges separated by a distance d. Find the approximate potential at points far from the dipole. (6)
 (ii) Find the electric field at a distance 'z' above the center of a circular loop of radius r, which carries a uniform line charge λ . (5)
 (iii) Given below the electric field variation find the odd one out.
 (1) $\mathbf{E} = c [xy\mathbf{i} + 2yz\mathbf{j} + 3xz\mathbf{k}]$
 (2) $\mathbf{E} = c [x^2\mathbf{i} - (2y^2 + z^2)\mathbf{j} + 2yz\mathbf{k}]$
 Find the potential for the possible field, using the origin as your reference point. (5)

12. (a) (i) Find the magnetic field at the centre of a square loop, which carries a steady current I. Let R be the distance from centre to side. Find the field at the centre of a n-sided polygon, carrying a steady current I. Again, let R be the distance from the centre to any side. Find the formula in the limit n (number of sides) tends to infinity. (8)
 (ii) Find the magnetic field a distance h above the center of a circular loop of radius R, which carries a steady current I. (8)

Or

- (b) (i) Derive the Ampere's law. (8)
 (ii) Derive the expressions which mutually relate Current density $\mathbf{J}$, Magnetic field $\mathbf{B}$ and Magnetic vector potential $\mathbf{A}$. (8)

13. (a) (i) Derive the expression for the energy of a point charge distribution. Three point charges -1 nC , 4 nC , and 3 nC are located at (0, 0, 0), (0, 0, 1) and (1, 0, 0) respectively. Find the energy in the system. (8)
 (ii) A small loop of wire (radius a) has a distance z above the center of a large loop (radius b). The planes of the two loops are parallel, and perpendicular to the common axis. Suppose current I flows in the big loop. Find the flux through the little loop. Find the mutual inductance. (8)

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- (b) (i) Write the Poisson's and Laplace's equations. (4)
- (ii) Discuss the magnetic boundary conditions. (6)
- (iii) Two concentric metal spherical shells of radii a and b are separated by weakly conducting material of conductivity σ . If they are maintained at a potential difference V , what current flows from one to the other? What is the resistance between the shells? Find the resistance if $b \gg a$. (6)
14. (a) (i) Explain Ampere's circuit law. (8)
- (ii) Derive Poynting's Theorem. (8)
- Or
- (b) (i) Describe the Maxwell's equations in differential and integral forms. (8)
- (ii) Write Faraday's law in differential and integral forms and explain Faraday's experiments. (8)
15. (a) (i) Derive the wave equations for Electric and Magnetic fields. (8)
- (ii) The electric field intensity of a linearly polarized uniform plane wave propagating in the $+z$ direction in seawater is $E = 100 \cos(10^8 t - \beta z)$ V/m at $z = 0$. The constitutive parameters of seawater are $\epsilon_r = 72$, $\mu_r = 1$, and conductivity $\sigma = 4$ S/m. Determine the attenuation constant, phase constant, intrinsic impedance, phase velocity, wavelength and skin depth. Also find the distance at which the amplitude of E is 1% of its value at $z = 0$. (8)
- Or
- (b) (i) Analyze the wave behaviour at boundaries under oblique incidence and derive the Brewster's angle. (12)
- (ii) Prove that a linearly polarized wave can be resolved into a right hand circularly polarized wave and a left hand circularly polarized wave of equal amplitude. (4)

Reg. No. :

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Question Paper Code : 91402

B.E./B.Tech DEGREE EXAMINATION, NOVEMBER/DECEMBER 2014

Fourth Semester

Electronics and Communication Engineering

EC 2253/ EC 13EC 1253/ 080290021/ 101 41 EC 404 — ELECTROMAGNETIC
FIELDS

(Regulation 2008/2010)

Time : Three hours

Maximum : 100 marks

Answer ALL questions

PART A — (10 × 2 = 20 marks)

1. State Stokes theorem and give its meaning.
2. A 15 nC point charge is at the origin in free space. Calculate V if point P is located at $P(-2, 3, -1)$ and $V = 0$ at $(6, 5, 4)$.
3. A current filament carrying 15 A in the z direction lies along the entire z axis. Find H in rectangular coordinates at $P(2, -4, 4)$.
4. What is Magnetic vector potential?
5. State point form of Ohm's law.
6. Find the magnetization in a magnetic material where $\mu = 1.8 \times 10^{-3}$ H/m and $H = 120$ A/m.
7. State Poynting vector.
8. Maxwell's Second Equation is based on a famous law. What is it? Substantiate.
9. What is Uniform Plane Wave?
10. Define Brewster angle.

PART B — (5 × 16 = 80 marks)

11. (a) (i) State Gauss law and explain its applications. (6)
- (ii) Three infinite uniform sheets of charge are located in free space as follows: 3 nC/m^2 at $z = -1$, 6 nC/m^2 at $z = 1$ and -8 nC/m^2 at $z = 4$. Find E at the points: $P_1(2,5,-5)$, $P_2(4,2,-3)$, $P_3(-1,-5,2)$ and $P_4(-2,4,5)$. (6)
- (iii) Point charges of 50 nC each are located at $A(1,0,0)$, $B(-1,0,0)$, $C(0,1,0)$ and $D(0,-1,0)$ in free space. Find the total force on the charge at A . (4)

Or

- (b) (i) Define Curl, Divergence and Gradient and state their meanings. (6)
- (ii) Find the potential due to an electric dipole. (6)
- (iii) Two uniform line charges, 8 nC/m each, are located at $x = 1$, $z = 2$ and at $x = -1$, $y = 2$ in free space. If the potential at the origin is 100 V find V at $P(4,1,0)$. (4)
12. (a) (i) Find H in rectangular components at $P(2,3,4)$ if there is a current filament on the z axis carrying 8 mA in the z direction. Repeat if the filament is located at $x = -1$ and $y = 2$. Find H if both filaments are present. (6)
- (ii) State Ampere's Circuital law and explain its applications. (6)
- (iii) A filamentary conductor is formed into an equilateral triangle with sides of length l carrying current I . Find the magnetic field intensity at the center of the triangle. (4)

Or

- (b) (i) State Lorentz force equation for a moving charge and explain its applications. (6)
- (ii) Derive the expression for Torque on a loop carrying a current I . (10)
13. (a) (i) State and prove the boundary conditions for static magnetic field and static electric field. (10)
- (ii) Derive the expression for electrostatic energy density. (6)

Or

- (b) (i) Derive the Capacitance of a parallel plate capacitor. (4)
- (ii) Calculate the self-inductances of and the mutual inductances between two coaxial solenoids R_1 and R_2 , $R_2 > R_1$, carrying currents I_1 and I_2 with n_1 and n_2 turns/m respectively. (6)
- (iii) Derive the expression for energy density in magnetic fields. (6)

14. (a) (i) Derive Maxwell's equations from basic principles. (10)
(ii) Derive the expression for power flow in a co-axial cable. (6)

Or

- (b) (i) Derive the expression for Poynting vector. (10)
(ii) Why is Ampere's circuital law modified? How is it modified? Substantiate. (6)
15. (a) (i) Derive Wave Equation from Maxwell's Equations. (8)
(ii) Describe the concept of Plane Wave propagation in good conductors. (8)

Or

- (b) Explain with relevant expressions, the concept of reflection of plane waves by a perfect dielectric at both normal and oblique incidence.

Reg. No. :

Question Paper Code : 71449

B.E./B.Tech. DEGREE EXAMINATION, APRIL/MAY 2015.

Fourth Semester

Electronics and Communication Engineering

EC 2253/EC 43/EC 1253/080290021/10144 EC 401 — ELECTROMAGNETIC
FIELDS

(Regulation 2008/2010)

Time : Three hours

Maximum : 100 marks

Answer ALL questions.

PART A — (10 × 2 = 20 marks)

1. State Stokes theorem.
2. What is the relationship between electric scalar potential and electric field intensity?
3. What is magnetic vector potential?
4. State Lorentz force equation for a moving charge.
5. State Poisson's and Laplace's equations.
6. What is mutual inductance?
7. What is displacement current?
8. Define Poynting Vector.
9. What is skin effect?
10. What is Brewster angle?

PART B — (5 × 16 = 80 marks)

11. (a) (i) A charge $Q_1 = 3 \times 10^{-4}$ C is at a point M(1,2,4) and a second charge $Q_2 = -10^{-4}$ C located at a point N(2,0,10) in vacuum. Find the force exerted on Q_2 by Q_1 . (4)
- (ii) Infinite uniform line charges of 5 nC/m lie along the x and y axes in free space. Find E at $P_A(0, 0, 4)$ and at $P_B(0, 3, 4)$. (4)
- (iii) Derive an expression for Electric field on the axis of a uniformly charged circular disc. (8)

Or

- (b) (i) Define divergence and curl. (4)
- (ii) Derive an expression for potential due to electric dipole. (6)
- (iii) State Gauss law and prove it. (6)
12. (a) (i) Find H in rectangular co-ordinates at P(2,3,4) if there is a current filament on the z axis carrying 8 mA in the z_+ direction. (4)
- (ii) Express Biot-Savart Law in vector form and describe it. (4)
- (iii) State Ampere's circuital law and discuss about any two simple applications of it. (8)

Or

- (b) (i) Derive an expression for Torque on a loop carrying a current I. (12)
- (ii) Define magnetic flux density and magnetic moment. (4)
13. (a) (i) Derive the boundary conditions for electric fields. (8)
- (ii) Derive the expressions for electrostatic energy and energy density. (8)

Or

- (b) (i) State continuity equation for current and point form of ohm's law. (4)
- (ii) Discuss in detail the nature of magnetic materials. (6)
- (iii) A solenoid is 50cm long, 2 cm in diameter and contains 1500 turns. The cylindrical core has a diameter of 2 cm and a relative permeability of 75. This coil is coaxial with a second solenoid 50cm long, 3 cm in diameter and having 1200 turns. Calculate the inductance for the inner solenoid; find inductance of the outer solenoid; determine mutual inductance between the two solenoids. (6)

14. (a) (i) From basic principles, derive Maxwell's four equations in integral form and differential form. (12)
 (ii) State the modified form of Ampere's circuital law. Why was it modified? Justify. (4)

Or

- (b) (i) Derive expressions for Instantaneous, Average and Complex Poynting Vector. (12)
 (ii) Interpret $\mathbf{E} \times \mathbf{H}$. (4)
15. (a) (i) Derive Wave equation from Maxwell's equations. (6)
 (ii) What is a Uniform Plane Wave? Derive the relation between $\mathbf{E}$ and $\mathbf{H}$ in a Uniform Plane Wave. (10)

Or

- (b) (i) Discuss in detail with relevant expressions and figures, the concepts involved in reflection of plane waves by a perfect conductor at normal incidence. (10)
 (ii) A uniform plane electromagnetic wave is incident normally upon a sheet of dielectric material, which has the following constants: $\epsilon = 4\epsilon_0$, $\mu = \mu_0$, $\sigma = 0$. If the sheet is 2cm thick and the amplitude of the electric field strength of the incident wave is 100mv/m, determine the electric field strength of the wave after passing through the sheet, if the frequency is 3000 MHz; if the frequency is 30 Hz. Comment on the results. (6)

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Question Paper Code : 57286

B.E./B.Tech. DEGREE EXAMINATION, MAY/JUNE 2016

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Fourth Semester

Electronics and Communication Engineering

EC 6403 – ELECTROMAGNETIC FIELDS

(Regulations 2013)

Time : Three Hours

Maximum : 100 Marks

Answer ALL questions.

PART – A ($10 \times 2 = 20$ Marks)

1. State divergence theorem.
2. Define electric dipole.
3. Write the Laplace's equations in all the three coordinates.
4. What is dielectric polarization ?
5. Define magnetic vector and scalar potential.
6. A current of 3A flowing through an inductor of 100mH. What is the energy stored in inductor ?
7. Mention the force between two current elements.
8. Differentiate diamagnetic, paramagnetic and ferromagnetic material.
9. State Faraday's law of induction.
10. What is Poynting Vector ?

PART – B (5 × 16 = 80 Marks)

11. (a) Define the potential difference and electric field. Give the relation between potential and field intensity. Also Derive an expression for potential due to infinite uniformly charged line and also derive potential due to electric dipole. (16)

OR

- (b) (i) State and prove Gauss law and explain any one of applications of Gauss law. (8)
- (ii) Given two vectors $\vec{A} = 3\hat{a}_x + 4\hat{a}_y - 5\hat{a}_z$ and $\vec{B} = -6\hat{a}_x + 2\hat{a}_y + 45\hat{a}_z$, determine the unit vector normal to the plane containing the vectors $\vec{A}$ and $\vec{B}$. (8)
12. (a) (i) Derive the relationship between polarization and electric field intensity. (8)
- (ii) Derive the capacitance of a spherical capacitor. (8)

OR

- (b) (i) Derive the boundary conditions of the tangential and normal components of electric field at the interface of two mediums with dielectrics. (10)
- (ii) If two parallel plates of area 4 m^2 are separated by a distance 6 mm, find the capacitance between these 2 plates. If a rubber sheet of 4 mm thick with $\epsilon_r = 2.4$ is introduced in between the plates leaving a gap of 1 mm on both sides, determine the capacitance. (6)

13. (a) State Biot-Savart's law. Derive the expressions for magnetic field intensity and magnetic flux density at the centre of the square current loop of side l . Then determine the same for square loop of sides 5m carrying current of 10 A. (16)

OR

- (b) Derive an expression for magnetic field due to an infinitely long coaxial cable. (16)
14. (a) (i) Derive the expression for force on a moving charge in a magnetic field and Lorentz force equation. (8)
- (ii) Derive the inductance of a toroid. (8)

OR

- (b) (i) Derive an expression for inductance of a solenoid. Calculate the inductance of solenoid, 8 cm in length, 2 cm in radius, having $\mu_r = 100$ and 1000 turns. (8)
- (ii) Give the comparison between magnetic and electric circuits. (8)

15. (a) Derive the Maxwell's equation in differential and integral forms. (16)

OR

- (b) Starting from Maxwell's equation, derive homogeneous vector Helmholtz's equation in phasorform. (16)

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Question Paper Code : 80338

B.E./B.Tech. DEGREE EXAMINATION, NOVEMBER/DECEMBER 2016.

Fourth Semester

Electronics and Communication Engineering

EC 6403 – ELECTROMAGNETIC FIELDS

(Regulations 2013)

Time : Three hours

Maximum : 100 marks

Answer ALL questions.

PART A — (10 × 2 = 20 marks)

1. State coulombs law.
2. What is an electric potential? Write expression for potential due to an electric dipole.
3. Define resistance of a conductor.
4. Give Laplace's and Poisson's equations.
5. State Ampere's circuital law.
6. What is vector magnetic potential?
7. Calculate the mutual inductance of two inductively tightly coupled coils with self-inductance of 25 mH and 100 mH.
8. Give the expression for Lorentz force equation.
9. Define Phase velocity.
10. Find the displacement current density for field $E = 300 \sin 10^9 t \text{ V/m}$.

PART B — (5 × 16 = 80 marks)

11. (a) (i) State and Prove Stokes theorem. (8)
(ii) Derive the expression for energy and energy density in static electric fields. (8)

Or

- (b) (i) A circular disc of radius ' a ' meter is charged uniformly with a charge of ρ c/m. Find the electric field intensity at a point h meter from the disc along its axis. (10)
- (ii) Explain the concept of superposition principle of electric field intensity. (6)

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12. (a) Derive an expression for capacitance of a coaxial cable. (16)

Or

- (b) (i) Derive an expression for Polarization 'P'. (4)
 (ii) State and explain the electric boundary conditions between two dielectrics materials. (12)
13. (a) From Biot-Savart's law obtain expression for magnetic field intensity and vector potential at a point P and distance ' R ' from infinitely long straight current carrying conductor. (16)

Or

- (b) (i) Consider two identical circular current loops of radius 3 m and opposite current 20 Amps are in parallel planes, separated on their common axis by 10 m. Find the magnetic field intensity at a point midway between the two loops. (8)
 (ii) State Biot-Savart's law. Find the magnetic Field intensity at the origin due to current element $Id\vec{l} = 3\pi(\hat{a}_x + 2\hat{a}_y + 3\hat{a}_z) \mu A.m$ at (3, 4, 5) in free space. (8)
14. (a) (i) A charged particle with velocity $\vec{u}$ is moving in a medium containing uniform field $\vec{E} = E\hat{a}_x V/m$ and $\vec{B} = B\hat{a}_y Wb/m^2$. What should $\vec{u}$ be so that the particle experiences no net force on it? (8)
 (ii) State and derive the magnetic boundary conditions between the two magnetic mediums. (8)

Or

- (b) Derive the expression for inductance and magnetic flux density inside the solenoid. Calculate the inductance of the solenoid and energy stored when a current of 8 A flowing through the solenoid of 2m long, 10 cm diameter and 4000 turns. (16)
15. (a) (i) State and prove Poynting's theorem and give its physical interpretation. (8)
 (ii) Derive Maxwell's equations for time varying fields. (8)

Or

- (b) Derive the wave equation starting from Maxwell's equation for free space. (16)

for all subject Notes & QP's

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Reg. No. :

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Question Paper Code : 71730

B.E./B.Tech. DEGREE EXAMINATION, APRIL/MAY 2017.

Fourth Semester

Electronics and Communication Engineering

EC 6403 — ELECTROMAGNETIC FIELDS

(Regulations 2013)

Time : Three hours

Maximum : 100 marks

Answer ALL questions.

PART A — (10 × 2 = 20 marks)

1. Define gradient of a scalar field.
2. State divergence theorem.
3. Write the equation for energy stored in electrostatic field in terms of field quantities.
4. What is the practical application of method of images?
5. Define capacitance and capacitors.
6. An infinitesimal length of wire is located at (1, 0, 0) and carries a current 2A in the direction of unit vector $\hat{a}_z$. Find the magnetic flux density B due to the current element at the field point (0, 2, 2).
7. Define skin depth.
8. Define dielectric strength.
9. Differentiate conduction current and displacement current.
10. List any two properties of uniform plane waves.

PART B — (5 × 13 = 65 marks)

11. (a) Given $D = 2rz^2\hat{a}_r + r\cos^2\phi\hat{a}_z$. Prove divergence theorem. (13)
- Or
- (b) (i) Using gauss law find the electric field intensity for the uniformly charged sphere of radius 'a' find the E everywhere. (8)
- (ii) Derive the equation for scalar electric potential. (5)

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12. (a) (i) Derive vector magnetic potential from BiotSavart law. (8)
 (ii) Classify the materials based on magnetic properties. (5)

Or

- (b) (i) Find the magnetic flux density for the infinite current sheet in the xy plane with current density $K = K_y \hat{a}_y$ A/m current. (7)
 (ii) Derive the equation to find the force between the two current elements. (6)

13. (a) Derive the boundary condition for the E-field and H-field in the interface between dielectric and free space.

Or

- (b) (i) Find the capacitance for a coaxial capacitor with inner radius ' a ' and outer radius ' b ' with length L . (7)
 (ii) Derive the equation for the magnetization for the materials and show that $J_b = \nabla \times m$ and $K_b = m \times \hat{a}_n$. (6)

14. (a) From the basic laws derive the time varying Maxwell's equation and explain the significance of each equation in detail. (13)

Or

- (b) (i) State and derive poynting theorem. (8)
 (ii) Explain the transformer emf using Faraday's law. (5)

15. (a) Starting from Maxwell's equation derive the equation for E field in the form of wave in free space. (13)

Or

- (b) Explain the condition and propagation of uniform plane waves in good conductors and derive the wave constants. (13)

PART C — (1 × 15 = 15 marks)

16. (a) With relevant examples explain in detail the practice application of electromagnetic fields. (15)

Or

- (b) (i) Find the expression of induction for the co-axial. (8)
 (ii) Propose the salient points to be noted when the boundary conditions are applied. (7)

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Question Paper Code : 50439

B.E./B.Tech. DEGREE EXAMINATION, NOVEMBER/DECEMBER 2017

Fourth Semester

Electronics and Communication Engineering

EC 6403 : ELECTROMAGNETIC FIELDS

(Regulations 2013)

Time : Three Hours

Maximum : 100 Marks

Answer ALL questions

PART - A

(10×2=20 Marks)

1. State Gauss Law.
2. State stokes theorem.
3. What is polarization ?
4. Define skin depth.
5. State amperes circuital law.
6. A long straight wire carries a current $I = 10 \text{ mA}$. At what distance is the magnetic field intensity is 15 A/m ?
7. What is the inductance of a toroid for the coil of N turns ?
8. Write the Lorentz force equation for a moving charge.
9. State Faradays law.
10. What is the importance of Poynting vector ?

PART - B

(5×13=65 Marks)

11. a) Find the electric field due to infinite long conductor and infinite sheet of charge using Gauss law. (13)
- (OR)
- b) Derive the energy stored in electrostatic field in terms of field quantities. (13)

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50439



12. a) A cylindrical capacitor consists of an inner conductor of radius 'a' and an outer conductor whose inner radius is 'b'. The space between the conductors is filled with a dielectric permittivity ϵ_r and length of the capacitor is L. Find the value of the capacitance. (13)

(OR)

- b) i) State the relationship between polarization and electric field intensity. (7)
 ii) Write down the general procedure for solving Poisson's and Laplace's equation. (6)

13. a) Derive a general expression for the magnetic flux density B, at any point along the axis of a long solenoid. (13)

(OR)

- b) Using Biot-Savart's law, determine the magnetic field intensity due to a straight current carrying filamentary conductor of finite length AB. (13)

14. a) Derive the boundary conditions for magnetostatic fields at the interface of two different medium with permeability μ_1 and μ_2 . (13)

(OR)

- b) Planes $Z = 0$ and $Z = 4$ carry current $K = -10 \mathbf{a}_x$ A/m and $K = 10 \mathbf{a}_x$ A/m, respectively. Determine H at (1, 1, 1) and (0, -3, 10). (13)

15. a) Derive the Maxwell's equation in point and integral form. (13)

(OR)

- b) Deduce the Poynting's theorem from Maxwells equation and find the total time average power, crossing a given surface S. (13)

PART - C

(1×15=15 Marks)

16. a) In a medium characterized by $\sigma = 0$, $\mu = \mu_0$, $\epsilon = 4\epsilon_0$ and $\mathbf{E} = 20 \sin(10^8 t - \beta z) \mathbf{a}_y$ V/m. Calculate β and H.

(OR)

- b) A parallel-plate capacitor with plate area of 5 cm^2 and plate separation of 3 mm has a voltage $50 \sin 10^3 t$ V applied to its plates. Calculate the displacement current assuming $\epsilon = 2\epsilon_0$.

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Question Paper Code : 40957

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B.E./B.Tech. DEGREE EXAMINATION, APRIL/MAY 2018

Fourth Semester

Electronics and Communication Engineering

EC 6403 – ELECTROMAGNETIC FIELDS

(Regulations 2013)

Time : Three Hours

Maximum : 100 Marks

Answer ALL questions

PART – A

(10×2=20 Marks)

1. State divergence theorem.
2. Write the different coordinate systems.
3. Two capacitances C_1 and C_2 are connected in series. Find the equivalent total capacitance.
4. What is current density ?
5. What is vector magnetic potential ?
6. State Ampere circuital law.
7. Define – dielectric strength.
8. What is ferromagnetic material ?
9. What is electromotive force ?
10. Define Poynting's theorem.

PART – B

(5×13=65 Marks)

11. a) i) State and prove Stokes theorem. (6)
 ii) Derive Electric field intensity due to line charge. (7)
 (OR)
- b) i) Derive the equation for potential difference to move a point charge in electric field. (7)
 ii) Derive the Electric field due to electric dipole. (6)

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40957



12. a) Derive an expression for energy stored in the electrostatic field of a section of a coaxial cable. (13)

(OR)

- b) Derive the electric field boundary condition, when a wave travels between two different dielectrics medium. (13)

13. a) State Biot-Savart's law and derive the expressions for magnetic field intensity, of a straight current carrying conductor.

(OR)

- b) Derive the magnetic field intensity of a circular current carrying conductor.

14. a) Derive the inductance of toroid and solenoid.

(OR)

- b) i) Derive the equation which relates magnetization and permeability. (8)

- ii) Explain the different types of magnetic materials. (5)

15. a) Derive the Poynting theorem equation from Maxwell's curl equation.

(OR)

- b) Derive the Maxwell's equations in Differential form and integral form.

PART – C

(1×15=15 Marks)

16. a) Apply Lorentz force equation, to derive the force on a differential current element.

(OR)

- b) Illustrate with an example, to apply Poisson's and Laplace equation.
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Question Paper Code : 20413



DEGREE EXAMINATION, NOVEMBER/DECEMBER 2018.

Fourth Semester

Electronics and Communication Engineering

EC 6403 — ELECTROMAGNETIC FIELDS

(Regulations 2013)

(Common to PTEC 6403 — Electromagnetic Fields for B.E. (Part-Time) Third Semester + Electronics and Communication Engineering — Regulations 2014)

Time : Three hours

Maximum : 100 marks

Answer ALL questions.

PART A — (10 × 2 = 20 marks)

1. State divergence theorem.
2. Specify the applications of Gauss law.
3. Define magneto static vector potential.
4. Mention the Laplace equation in electromagnetic field.
5. What is meant by Biot-Savart Law?
6. State stokes theorem.
7. Outline the Amperes law of force between current carrying conductors.
8. Determine the force and torque in terms of mutual inductance.
9. Relate electrostatic and Magneto static models.
10. Outline the fundamental postulate for Electromagnetic Induction.

PART B — (5 × 13 = 65 marks)

11. (a) Illustrate in detail about the coulomb's law in electric fields. (13)

Or

- (b) Determine the electric field intensity of an infinitely long, straight line charge of a uniform density ρ in air. (13)

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12. (a) Derive the boundary conditions for electrostatic fields. (13)

Or

- (b) A parallel plate capacitor consists of two parallel conducting plates of area S separated by a uniform distance d . The space between the plates is filled with a dielectric of a constant permittivity, ϵ . Determine the capacitance. (13)

13. (a) An infinitely long, straight conductor with a circular cross section of radius b carries a steady current I . Determine the magnetic flux density both inside and outside the conductor. (13)

Or

- (b) Derive the vector magnetic potential. (13)

14. (a) Find the inductance per unit length of a very long solenoid with air core having ' n ' turns per unit length. (13)

Or

- (b) Determine the force per unit length between two infinitely long parallel conducting wires carrying currents I_1 and I_2 in the same direction. The wires are separated by a distance d . (13)

15. (a) Derive the integral form of Maxwell's equations. (13)

Or

- (b) A circular loop of N turns of conducting wire lies in the xy -plane with its center at the origin of a magnetic field specified at the origin of a magnetic field specified by $B = a_z B_0 \cos(\pi/2b) \sin \omega t$, where ' b ' is the radius of the loop and ω is the angular frequency. Find the emf induced in the loop. (13)

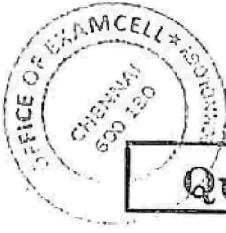
PART C — (1 × 15 = 15 marks)

16. (a) Express $3 \cos \omega t - 4 \sin \omega t$, as first (i) $A_1 \cos(\omega t + \theta_1)$ and then (ii) $A_2 \sin(\omega t + \theta_2)$. Determine A_1, θ_1, A_2 and θ_2 . (15)

Or

- (b) A rectangular loop in the xy -plane with sides b_1 and b_2 carrying a current I lies in a uniform magnetic field $B = a_x B_x + a_y B_y + a_z B_z$. Determine the force and torque on the loop. (15)

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Question Paper Code : 52912

B.E./B.Tech. DEGREE EXAMINATION, APRIL/MAY 2019.

Fourth Semester

Electronics and Communication Engineering

EC 6403 – ELECTROMAGNETIC FIELDS

(Regulation 2013)

(Common to PTEC 6403 – Electromagnetic Fields for B.E. Part – time for Third Semester – Electronics and Communication Engineering – Regulation 2014)

Time : Three hours

Maximum : 100 marks

Answer ALL questions.

PART A — (10 × 2 = 20 marks)

1. If a vector $\vec{A} = \vec{a}_x + 2\vec{a}_y + 3\vec{a}_z$, find its magnitude.
2. In Cartesian coordinates, a point is described by $P(1, 2, 4)$ Identify the orthogonal planes whose intersection give this point.
3. State stokes theorem.
4. Give the relationship between potential gradient and electric field.
5. Define dielectric strength.
6. Why water has much greater dielectric constant than mica?
7. State amperes circuital law.
8. Define magnetic moment.
9. Write down the constitutive relations.
10. State Maxwells Third equation.

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PART B — ($5 \times 13 = 65$ marks)

11. (a) Obtain the differential length, volume and surface elements in cylindrical coordinate system.

Or

- (b) An infinitely long line charge of uniform density ρ_L C/m is placed along Z – axis. Find the expression for electric field intensity at a point in Y – axis, which is 'a' meters away from 'Z' axis.

12. (a) State Biot Savart Law and its expression.

Or

- (b) Elaborate the applications of Poisson's and Laplace's equations in detail.

13. (a) Compute the magnetic field of a long straight wire that has a circular loop with a radius of 0.05m. 2amp is the reading of the current flowing through this closed loop.

Or

- (b) Derive Maxwells equation in point form and in integral form.

14. (a) Derive the Poynting vector from Maxwells equations and give its significance.

Or

- (b) Derive an expression of inductance of toroid and solenoid.

15. (a) Find the expression for magnetic field intensity due to an infinite long straight conductor carrying a current of I amperes at a point which has the distance of 'a' from the conductor.

Or

- (b) Derive the vector wave equation and give its physical interpretation.

PART C — ($1 \times 15 = 15$ marks)

16. (a) Summarize the concept of transformer and motional emf.

Or

- (b) Derive an expression of self-inductance and mutual inductance.

Reg. No. :

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Question Paper Code : 80120

B.E./B.Tech. DEGREE EXAMINATIONS, APRIL/MAY 2019.

Fourth Semester

Electronics and Communication Engineering

EC 8451 – ELECTROMAGNETIC FIELDS

(Common to Electronics and Telecommunication Engineering)

(Regulation 2017)

Time : Three hours

Maximum : 100 marks

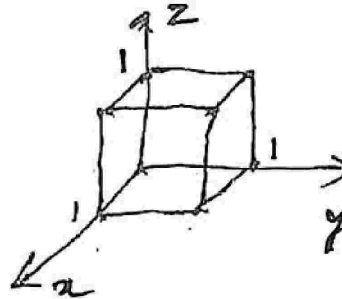
Answer ALL questions.

PART A — ($10 \times 2 = 20$ marks)

1. Write Stoke's theorem in integral form.
2. Define infinitesimal volume element in spherical polar coordinates.
3. Write coulomb's law.
4. Find the energy of a uniformly charged spherical shell of total charge q with a radius R .
5. Write Lorentz force equation.
6. Find the magnetic field a distance s from a long straight wire carrying a steady current I .
7. What is meant by displacement current?
8. Write electromagnetic boundary conditions.
9. What is meant by Brewster's angle?
10. Define phase velocity and group velocity.

PART B — (5 × 13 = 65 marks)

11. (a) Check the divergence theorem using the function $V = y^2\hat{i} + (2xy + z^2)\hat{j} + (2yz)\hat{k}$ and the unit cube situated at the origin.

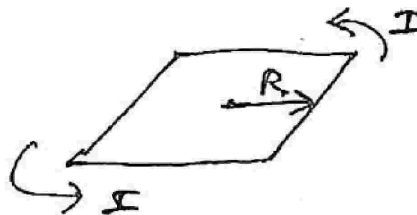


Or

- (b) Write the infinite small displacement, surface and volume elements in spherical and cylindrical coordinates.
12. (a) Find the electric field a distance Z above the center of a square loop of side ' a ' carrying uniform line charge λ .

Or

- (b) Derive the expressions for the energy of a (i) point charge distribution (ii) continuous charge distributions.
13. (a) Find the magnetic field at the center of a square loop, which carries a steady current I . Let ' R ' be the distance from center to side (fig.). Find the field at the center of an n -sided polygon, carrying a steady current I . Again, let R be the distance from the center to any side. Find the formula in the limit n (number of sides) tends to infinity.



Or

- (b) Define (i) the mutual inductance between two circuits, and (ii) self inductance of a single coil. Also explain how the self inductance of a wire-wound inductor depends on its number of turns.

14. (a) Write Maxwell's equations in differential form and integral form. Examine them and give its physical interpretation.

Or

- (b) Derive wave equations for electric and magnetic fields.

15. (a) Derive Poynting theorem.

Or

- (b) Analyse the wave reflection and transmission at normal incidence at the boundary between two linear media.

PART C — (1 × 15 = 15 marks)

16. (a) A 1.8 KHz wave propagates in a medium characterized by $\mu_r = 1.6$, $\epsilon_r = 25$ and conductivity $\sigma = 2.5$ s/m. The electric field intensity in the region is given by $\vec{E} = 0.1e^{-\alpha z} \cos(2\pi ft - \beta z) \hat{i}$ V/m. Determine the attenuation constant, propagation constant, intrinsic impedance, phase velocity, skin depth, and wave length of the wave.

Or

- (b) Two grounded conducting planes ($y=0$ and $x=0$) are intersecting at 90° . A charge of 100 nC is placed at (3, 4, 0). Find the electric potential and electric field intensity at (3, 5, 0).

12. a) Explain boundary conditions for electric fields between conductor and free space.

(OR)

- b) Derive capacitance of a coaxial cable and for a spherical capacitor.

13. a) Derive magnetic field intensity for a Co-axial Cable.

(OR)

- b) i) Determine magnetic field intensity due to infinitely long conductor using Ampere's Law. (7)

- ii) Given the vector magnetic potential, $\vec{A} = \frac{10}{x^2 + y^2 + z^2} \vec{a}_x$, obtain the magnetic flux density. (6)

14. a) i) Derive the inductance of a coaxial cable. (7)

- ii) Derive the inductance of a solenoid. (6)

(OR)

- b) Explain and derive the magnetic field boundary condition for two different dielectrics.

15. a) Derive displacement current from circuital analysis and from Ampere circuital law.

(OR)

- b) Derive and explain Maxwell's equations both in integral and point forms.

PART - C

(1×15=15 Marks)

16. a) A point charge $Q = 18 \text{ nC}$ has a velocity of $5 \times 10^6 \text{ m/s}$ in the direction

$$\vec{a}_v = 0.6 \vec{a}_x + 0.75 \vec{a}_y + 0.3 \vec{a}_z$$

Calculate the magnitude of the force exerted on the charge by

$$\vec{E} = -3\vec{a}_x + 4\vec{a}_y + 6\vec{a}_z \text{ kv/m}$$

$$\vec{B} = -3\vec{a}_x + 4\vec{a}_y + 6\vec{a}_z \text{ MT}$$

(OR)

- b) Given a $60 \mu\text{C}$ point charge located at the origin. Find the electric flux passing through the closed surface defined by $\rho = 26 \text{ cm}$ and $z = \pm 26 \text{ cm}$.



Reg. No. :

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Question Paper Code : 90183
B.E./B.Tech. DEGREE EXAMINATIONS, NOVEMBER/DECEMBER 2019
Fourth Semester
Electronics and Communication Engineering
EC 8451 – ELECTROMAGNETIC FIELDS
(Common to Electronics and Telecommunication Engineering)
(Regulations 2017)

Time : Three Hours

Maximum : 100 Marks

Answer ALL questions

PART – A

(10×2=20 Marks)

1. State the fundamental theorem of divergence.
2. Write the volume integral to find the volume of a sphere of radius R.
3. An infinite plane carries a uniform surface charge σ . Find its electric field.
4. Write Poisson's equation.
5. Define Lorentz force law and give its expression.
6. Write divergence and curl of magnetic field.
7. State Faraday's law.
8. Find the ratio of conduction current density to displacement current density in terms of conductivity and angular frequency.
9. What is meant by group velocity?
10. Find the skin depth at frequency 1.6 MHz in Aluminum, where conductivity $\sigma = 38.2 \text{ Ms/m}$ and $\mu_r = 1$.

90183

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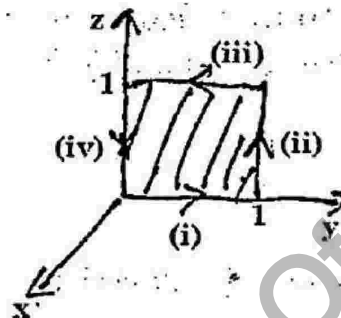
PART - B

(5×13=65 Marks)

11. a) Let $T = xy^2$ and take a point 'a' to be the origin (0, 0, 0) and 'b' the point (2, 1, 0). Check the fundamental theorem of gradients. (13)

(OR)

- b) Suppose $\mathbf{v} = (2xz + 3y^2)\hat{\mathbf{y}} + (4yz^2)\hat{\mathbf{z}}$. Check Stoke's theorem for the square surface shown in the figure. (13)



12. a) Define electric displacement and discuss electrostatic boundary conditions. (13)

(OR)

- b) State Coulomb's law and Gauss's law. Define electric potential. Write the relation between charge density, electric potential and electric field. (13)

13. a) Find the magnetic field at a distance 's' from a long straight wire carrying a steady current I by using Biot Savart law and Ampere's law. (13)

(OR)

- b) Derive the expression of energy in magnetic fields. (13)

14. a) Write Maxwell's equations in integral and differential forms. (13)

(OR)

- b) Derive the wave equations for electric and magnetic fields. (13)

15. a) Derive and state Poynting's theorem. (13)

(OR)

- b) Discuss the reflection and transmission of wave at normal incidence. (13)



PART - C

(1×15=15 Marks)

16. a) The electric field intensity of a linearly polarized uniform plane wave propagating in the +z direction in seawater is $\vec{E} = \hat{i} 100 \cos(10^7 \pi t) \left(\frac{V}{m} \right)$ at $z = 0$. $\epsilon_r = 72$, $\mu_r = 1$ and $\sigma = 4$ (S/m). Determine the attenuation constant, phase constant, intrinsic impedance, phase velocity, wave length and skin depth. Find the distance at which the amplitude of $\vec{E}$ is 1% of its value at $z = 0$. (15)

(OR)

- b) Two long coaxial cylindrical metal tubes (inner radius a , outer b) stand vertically in a tank of dielectric oil (susceptibility χ_v) (mass density ρ). The inner one is maintained at potential V and the outer one is grounded. To what height does the oil rise in the space between the tubes? (15)



Reg. No. :

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Question Paper Code : X10361

B.E./B.Tech. DEGREE EXAMINATIONS, NOVEMBER/DECEMBER 2020
AND APRIL/MAY 2021

Fourth Semester

Electronics and Communication Engineering

EC 8451 – ELECTROMAGNETIC FIELDS

(Common to : Electronics and Telecommunication Engineering)

(Regulations 2017)

Time : Three Hours

Maximum : 100 Marks

Answer ALL questions

PART – A

(10×2=20 Marks)

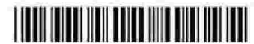
1. State Stokes theorem.
2. State Helmholtz's theorem.
3. State Coulomb's law for electric field.
4. Two point charges -1nC and 3nC are located at $(0, 0, 0)$, and $(1, 0, 0)$ respectively. Find the Electric Field Intensity of the system.
5. A current density $\mathbf{K} = 20a^x \text{ A/m}$ flows in the $y = 0$ plane through the region $-5 < z < 5 \text{ m}$ and $-\infty < x < \infty$. Find $\hat{\mathbf{H}}$ at $P(0, 10, 0)$ in free space.
6. State Biot Savart's law.
7. What is displacement current ?
8. State Faraday's law.
9. Define Poynting vector.
10. A sinusoidal electrical intensity of amplitude 250 V/m and frequency 1 GHz exists in a lossy dielectric medium that has a relative permittivity of 2.5 and loss tangent of 0.001 . Find the effective conductivity of the lossy medium.

PART – B

(5×13=65 Marks)

11. a) i) Give the Cartesian coordinates of the point $C(\rho = 4, \phi = -65^\circ, z = 2)$. (4)
 ii) Give the cylindrical coordinates of the point $D(x = -3.1, y = 2.6, z = 3)$. (4)
 iii) Specify the distance from C to D. (5)
 (OR)

X10361



- b) i) Find the div D at the Point $P(2, 3, -1)$ if $D = (2xy - y^2)a_x + (x^2z - 2xy)a_y + x^2ya_z$ C/m². (6)
 ii) Given vectors $M = -10ax + 4ay - 8az$ and $N = 8ax + 7ay + -2az$.
 Find (a) Unit vector in direction $-M + 2N$ (b) Magnitude of $5az + N - 3M$. (7)
12. a) Find the electrostatic field intensity $\vec{E}$ produced at the point $P(1, 5, 4)$ in free space due to the following static charge distributions :
 i) A uniform line charge with a charge density of 150 nC/m is located along X-axis. (6)
 ii) A uniformly charged sheet with a charge density of 25 nC/m² is located in the plane $z = -1$. (7)
 (OR)
- b) Derive the boundary conditions of the normal and tangential components of electric field at the interface of dielectric and free space.
13. a) Consider an infinitely long coaxial transmission line and obtain the expressions for magnetic field intensity ($\hat{H}$) everywhere using Ampere's circuit law. (OR)
 b) Determine the necessary boundary conditions between two different magnetic media μ_1 and μ_2 .
14. a) Illustrate the inconsistency of Ampere's circuital law for time varying fields, and what is the remedial solution proposed by Maxwell ? Also give the differential form representation of all the four Maxwell's equations. (OR)
 b) i) Derive wave equation from Maxwell's equation. (6)
 ii) Write Faraday's law in differential and integral forms and explain Faraday's experiments. (7)
15. a) What is uniform plane wave ? Derive the relationship between E and H in a uniform plane wave. (OR)
 b) Derive expressions for instantaneous, average and complex pointing vectors.

PART – C

(1×15=15 Marks)

16. a) Analyze the wave behavior at boundaries under oblique incidence and derive the Brewster's angle. (OR)
 b) i) Determine the expression of self inductance for a solenoid having inner radius 'a' and outer radius 'b'. (7)
 ii) Calculate the self inductance of a solenoid having 500 turns about a cylindrical core of 2 cm radius in which $\mu_r = 50$ for $0 < \rho < 0.5$ cm. (8)

Reg. No. :

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Question Paper Code : 20470

B.E./B.Tech. DEGREE EXAMINATIONS, APRIL/MAY 2022.

Fourth Semester

Electronics and Communication Engineering

EC 8451 — ELECTROMAGNETIC FIELDS

(Common to Electronics and Telecommunication Engineering)

(Regulations 2017)

Time : Three hours

Maximum : 100 marks

Answer ALL questions.

PART A — (10 × 2 = 20 marks)

1. Transform the Cartesian coordinates $x = 2$, $y = 1$, $z = 3$ into spherical coordinates.
2. Prove that curl gradient is zero.
3. Write down the expression for electric field intensity due to various charge distributions.
4. Write Poisson and laplace equation for electric field.
5. State biot savart law.
6. Calculate force between two wires carrying current of 5 A and 10 A in the same direction are placed with their axis 5 cm apart.
7. Find the amplitude of displacement current density inside a capacitor where $\epsilon_r = 600$ and $D = 3 \times 10^{-6} \sin(6 \times 10^6 t - 3464x) a_z \text{ C/m}^2$.
8. State faraday's law.
9. State poynting theorem.
10. Write the relation between reflection coefficient and standing wave ratio.

PART B — (5 × 13 = 65 marks)

11. (a) Why coordinate systems are required? Explain in detail about various coordinates systems.

Or

- (b) State and prove divergence and stokes theorem.

12. (a) Derive the expression for electric field intensity due to infinite sheet of charge.

Or

- (b) Define electric dipole and derive an expression for potential of a electric Dipole.

13. (a) State amperes law. Derive expression for magnetic field intensity due to solenoid, toroid and coaxial cable using amperes law.

Or

- (b) Show that the inductance of the cable is $L = \mu l/2\pi \ln(b/a)$.

14. (a) Derive Maxwell equation in point form, integral form and phasor form/ Harmonically time varying field.

Or

- (b) Derive the expression for electromagnetic wave equation for free space.

15. (a) Derive the expression for velocity of a wave when the wave propagates in dielectric medium.

Or

- (b) Derive transmission and reflection coefficient for the plane waves that incident oblique on Dielectric boundary.

PART C — (1 × 15 = 15 marks)

16. (a) Four point charges of $5 \mu\text{C}$ are placed in free space at the point (2, 0, 0), (-2, 0, 0), (0, 2, 0), (0, -2, 0) m respectively. Determine force on point charge of $30 \mu\text{C}$ located at a point (0, 0, 2).

Or

- (b) A capacitor is composed of two parallel sheets separated by a sheet of insulating material 3 mm thick and of relative permittivity $\epsilon_r = 4$. The distance between plates is increased to allow the insertion of a second sheet 5 mm thick and of relevant permittivity ϵ_{r2} . If the capacitance so formed is $1/3$ times of original capacitance calculate ϵ_{r2} .

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Question Paper Code : 90473

B.E./B.Tech. DEGREE EXAMINATIONS, NOVEMBER/DECEMBER 2022.

Fourth Semester

Electronics and Communication Engineering

EC 8451 — ELECTROMAGNETIC FIELDS

(Common to Electronics and Telecommunication Engineering)

(Regulations 2017)

Time : Three hours

Maximum : 100 marks

Answer ALL questions.

PART A — (10 × 2 = 20 marks)

1. Define gradient of a scalar field.
2. A vector function, $F = (4y - c_1z) a_x + (c_2x - 7z) a_y - (c_3y + 5z) a_z$. Determine the constants C_1, C_2, C_3 if F is irrotational.
3. State Coulomb's law.
4. The electrostatic field intensity E is derivable as the negative gradient of a scalar electric potential V . Determine E at the point (2,3,1) if $V = V_0 e^{-2x} \sin(\pi y/4)$.
5. State Ampere's Circuital law.
6. Vector magnetic potential in free space is $A = (2x^2y - yz) a_x + (xy^2 - xz^2) a_y - (6xy + 2xz) a_z$ (Wb/m). Calculate magnetic flux density (B).
7. What is the difference between conduction current and displacement current?
8. Write the boundary conditions of electromagnetic fields at air – conductor interface.
9. Write the properties of Uniform Plane Wave for free space.
10. Find the intrinsic impedance of the medium with permittivity $\epsilon = 4\epsilon_0$ and permeability $\mu = \mu_0$.

PART B — (5 × 13 = 65 marks)

11. (a) (i) A vector in cylindrical coordinates is given as $A = a_r A_r + a_\phi A_\phi + a_z A_z$. Derive the relation between the components of cylindrical coordinates and cylindrical coordinates. (7)
- (ii) State and explain helmholtz's theorem. (6)

Or

- (b) Verify the divergence theorem for the vector function $A = x^2 a_x + xy a_y + yz a_z$ over a cube one unit on each side. The cube is situated in the first octant of the cartesian coordinate system with one corner at the origin.
12. (a) Determine the Electric field intensity (E) of an infinitely long, straight, line charge of a uniform density ρ_l in air medium.

Or

- (b) A spherical capacitor consists of an inner conducting sphere of radius R_i and outer conductor with a spherical inner wall of radius R_o . The space between the conductors is filled with a dielectric of permittivity ϵ . Determine the capacitance of the capacitor.
13. (a) Derive the boundary conditions for the static magnetic field at the interface of two different magnetic medium with permeability μ_1 and μ_2 .

Or

- (b) An air coaxial transmission line has a solid inner conductor of radius a_1 and a very thin outer conductor of inner radius b_1 . Determine inductance per unit length of the line.
14. (a) Derive the integral and point form of the maxwell's equations from gauss law, ampere's law and faraday's law.

Or

- (b) Derive the wave equations from maxwell's equations and solve it for free space conditions.

15. (a) Derive the expression for attenuation and phase constant for an electromagnetic wave propagating in good conductor and dielectric medium.

Or

- (b) Find the poynting vector on the surface of a long, straight conducting wire that carries a direct current I . The radius and conductivity of the conductor are b and σ respectively. Verify the poynting's theorem.

PART C — ($1 \times 15 = 15$ marks)

16. (a) (i) A long, straight copper wire with 12 mm diameter carries a steady current of 10 mA. Determine the magnetic flux density and magnetic field intensity at 3 mm and 5 cm from the centre of the wire ($\mu = \mu_0 = 4\pi \times 10^{-7} \text{ H/m}$). (8)
- (ii) A point charge of $Q = 40 \text{ nC}$ is located at the origin in cartesian coordinates. Find the electric flux density and electric field intensity at $(1, 3, -5)$. (7)

Or

- (b) (i) A copper sheet is used as cover for electromagnetic shielding applications. What is thickness of copper sheet is required at 100 MHz, 500 MHz, 3 GHz and 30 GHz for the electromagnetic shielding applications? ($\mu = \mu_0 = 4\pi \times 10^{-7} \text{ H/m}$, $\sigma = 5.8 \times 10^7 \text{ S/m}$). (8)
- (ii) The Electric field intensity in air medium is given by $E = 10 \sin(100\pi x) \cos(12\pi 10^9 t - \beta z) \alpha_y \text{ (V/m)}$. Find the magnetic field intensity H and β . (7)

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Question Paper Code : 70512

B.E./B.Tech. DEGREE EXAMINATIONS, NOVEMBER/DECEMBER 2023.

Fourth Semester

Electronics and Communication Engineering

EC 8451 – ELECTROMAGNETIC FIELDS

(Common to Electronics and Telecommunication Engineering)

(Regulations 2017)

Time : Three hours

Maximum : 100 marks

Answer ALL questions.

PART A — ($10 \times 2 = 20$ marks)

1. List the three co-ordinate systems.
2. State Stokes theorem.
3. State Coulomb's Law.
4. What is the relation between Electric Flux Density and Electric Field Intensity?
5. Write two applications of Biot-Savart law.
6. State Ampere's circuital law.
7. State Faraday's law of electromagnetic induction.
8. What is displacement current?
9. Outline the importance of Poynting vector.
10. What is the intrinsic impedance of air?

PART B — ($5 \times 13 = 65$ marks)

11. (a) Derive the electric field due to infinite line charge with charge density ρL . (13)

Or

- (b) For a vector field A , show explicitly that $\nabla \cdot \nabla \times A = 0$ that is, the divergence of curl of any vector field is zero. (13)

12. (a) Explain boundary conditions for electric fields between conductor and free space. (13)

Or

- (b) A cylindrical capacitor consists of an inner conductor of radius 'a' and an outer conductor with inner radius 'b'. The space between the conductors is filled with a dielectric permittivity ϵ_r and length of the capacitor is L. Find the value of the capacitance. (13)
13. (a) Using Biot-Savart's law, derive the magnetic field intensity on the axis of a circular loop of radius R carrying a steady current I. (13)

Or

- (b) Formulate the magnetic flux density around infinitely long straight conductor by magnetic vector potential. (13)
14. (a) Derive pointing vector in integral and point form from Maxwell's equation. (13)

Or

- (b) Derive displacement current from circuital analysis and from Ampere circuital law. (13)
15. (a) Derive the attenuation constant, phase constant, phase velocity and intrinsic impedance for free space, dielectric and perfect conductors. (13)

Or

- (b) Find the reflection and transmission coefficient of a plane wave in oblique incident at a boundary of two dielectric mediums. (13)

PART C — (1 × 15 = 15 marks)

16. (a) If the vector magnetic potential is given by

$$\vec{A} = \frac{10}{x^2 + y^2 + z^2} \alpha_x \text{ Obtain the magnetic flux density in vector form. (15)}$$

Or

- (b) If $V = \left[2x^2y + 20z - \frac{4}{x^2 + y^2} \right]$ volts. Evaluate $\vec{E}$ and $\vec{D}$ at point $P(6, -2, 5, 3)$. (7 + 8)

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Question Paper Code : 30143

B.E./B.Tech. DEGREE EXAMINATIONS, APRIL/MAY 2023.

Fourth Semester

Electronics and Communication Engineering

EC 3452 — ELECTROMAGNETIC FIELDS

(Common to : Electronics and Telecommunication Engineering)

(Regulations 2021)

Time : Three hours

Maximum : 100 marks

Answer ALL questions.

PART A — (10 × 2 = 20 marks)

1. What is divergence of a vector field?
2. Calculate the curl of gradient of the scalar field, $V = 3xy - yz$.
3. Define Gauss's Law.
4. What is the significance of Laplacian Operator?
5. Define Ampere's Law.
6. What is the significance of magnetic vector potential?
7. What is the displacement current?
8. What is the significance of Continuity Equation?
9. Define skin depth
10. "X-rays can penetrate the human body, but light cannot". Justify.

PART B — (5 × 13 = 65 marks)

11. (a) Explain different type of coordinate systems along with examples of their use.

Or

- (b) Explain Gradient, Divergence and Curl in detail along with examples.

(3+5+5=13)

12. (a) Determine the net electric flux leaving through closed surface defined by:
 $\rho = 4, 0 \leq z \leq 1$, if electric flux density is given by:
 $D = \rho^2 \cos^2 \phi \alpha_\rho + z \sin \phi \alpha_\phi \text{ C/m}^2$.

Or

- (b) Derive the expression for Electric Field Intensity, E at a distance, r due to a volume charge as sphere of radius, R carrying charge density, $\rho_v \text{ C/m}^3$. Consider
- (i) $r < R$ (6.5)
- (ii) $r > R$. (6.5)
13. (a) Find the magnetic field intensity, H and magnetic flux density, B at the centre of the circular loop carries a current of I Amperes in the clockwise direction at $z = 0$ plane. Assume the diameter of the circular loop is $2a$.

Or

- (b) For a current distribution in free space, magnetic vector potential is given by : $A = (2x^2z + yz)\alpha_x + (xy^2 - xz^3)\alpha_y - (6xyz - 2x^2y^2)\alpha_z \text{ Wb/m}$.

Calculate :

- (i) magnetic flux density, B and (5)
- (ii) the flux crossing through the surface described by $x = 1, 0 < y < 2, 0 < z < 2$. (8)
14. (a) Explain in detail Maxwell's equations in integral and differential form for time varying fields.

Or

- (b) Explain the phenomenon of EM wave propagation in free space using Maxwell's Equations.
15. (a) The electric field intensity of a uniform plane wave in an unknown medium is given by :

$$E(y, t) = 25 \sin(10^8 t - y) \alpha_z \text{ V/m}.$$

Calculate :

- (i) nature of the medium, (2)
- (ii) attenuation constant (α), (2)
- (iii) phase constant (β), (2)
- (iv) phase velocity (v_p) and (2)
- (v) derive the expression for magnetic field intensity, H . (5)

Or

- (b) In a certain lossy medium, an EM wave travels for a distance of 10 m where its amplitude decays to $1/e$ times of its initial value. If the phase shift for the same period is 60° , then Calculate the propagation constant of the medium.

PART C — (1 × 15 = 15 marks)

16. (a) Three infinite sheets with charge density of 18 nC/m^2 , 9 nC/m^2 and -24 nC/m^2 are located at $x = 4$, $y = -3$ and $z = 0$ respectively. Find the electric field intensity at

(i) $(8, 9, 6)$ and (7.5)

(ii) $(-2, -7, 1)$. (7.5)

Or

- (b) An EM wave travels from a free space to a dielectric medium with dielectric constant $(\epsilon_r) = 4$ and it incidents normally on the interface. If the electric field of incident wave in free space is given by :

$$E_i = E_0 \cos(\omega t - \beta z) \hat{a}_z \text{ V/m, where } \omega = 3 \times 10^8 \pi \text{ and } \beta = 10\pi.$$

Then, calculate the value of

(i) reflection coefficient (Γ_E) . (3)

(ii) transmission coefficient (τ_E) . (3)

(iii) the fraction of power transmitted into the dielectric medium and, (3)

(iv) derive the expression for electric field of the transmitted wave (E_T) . (6)

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Question Paper Code : 20930

B.E./B.Tech. DEGREE EXAMINATIONS, NOVEMBER/DECEMBER 2023.

Fourth Semester

Electronics and Communication Engineering

EC 3452 – ELECTROMAGNETIC FIELDS

(Common to Electronics and Telecommunication Engineering)

(Regulations 2021)

Time : Three hours

Maximum : 100 marks

Answer ALL questions.

PART A — ($10 \times 2 = 20$ marks)

1. Define scalar field.
2. State Divergence theorem.
3. Define electric flux density.
4. List two applications of Gauss's Law.
5. Differentiate static electric and magnetic field lines.
6. What is a torque?
7. State Faraday's law.
8. What is potential difference?
9. What is group velocity?
10. List four properties of good conductors.

PART B — ($5 \times 13 = 65$ marks)

11. (a) Given, $A = (\sin 2\phi) a_\phi$ in cylindrical co-ordinates. Find curl of A at $(2, \pi/4, 0)$. (13)

Or

- (b) Given the two points, $A(x=2, y=3, z=1)$ and $B(r=2, \theta=20^\circ, \phi=220^\circ)$. Find

(i) Spherical co-ordinates of A (7)

(ii) Cartesian co-ordinate of B . (6)

12. (a) A charge is distributed on x axis of Cartesian system having a line charge density of $3x^2 \mu C/m$. Find the total charge over the length of 10 m. (13)

Or

- (b) If $V = x - y + xy + 2z$ V, Find
 (i) E at (2, 2, 1) (7)
 (ii) Energy stored in a cube of side 1m centered at the origin. (6)

13. (a) Obtain the expression for magnetic field intensity on an axis of a circular ring. (13)

Or

- (b) Find the magnetic field intensity at a point P, due to a finite straight conductor, carrying a current I. (13)

14. (a) Derive wave equation, and explain the properties of uniform plane waves in free space. (13)

Or

- (b) Derive and explain, Maxwell's equations both in integral and point forms. (13)

15. (a) Derive pointing vector in integral and point form from Maxwell's equation. (13)

Or

- (b) Explain the reflection of plane wave by conducting medium, under normal incidence. (13)

PART C — (1 × 15 = 15 marks)

16. (a) (i) A conductor 6m long, lies along Z direction with a current of 2A in a_z direction. Find the force experienced by conductor if $B = 0.08 a_x(T)$. (8)

- (ii) Find the maximum torque on an 85 turns, rectangular coil with dimension $(0.2 \times 0.3)m$, carrying current of 5 Amps in a field $B = 6.5T$. (7)

Or

- (b) (i) Position of a point P is $(8, 120^\circ, 330^\circ)$. Express in cylindrical co-ordinate system. (7)

- (ii) Determine the divergence of the vector field $\vec{P} = x^2 y z \vec{a}_x + x z \vec{a}_z$. (8)

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Question Paper Code : 50963

B.E./B.Tech. DEGREE EXAMINATIONS, APRIL/MAY 2024.

Fourth Semester

Electronics and Communication Engineering

EC 3452 – ELECTROMAGNETIC FIELDS

(Common to Electronics and Telecommunication Engineering)

(Regulations 2021)

Time : Three hours

Maximum : 100 marks

Answer ALL questions.

PART A — (10 × 2 = 20 marks)

1. Transform the vector $\vec{Q} = \frac{\sqrt{x^2 + y^2}}{\sqrt{x^2 + y^2 + z^2}} \vec{a}_x - \frac{yz}{\sqrt{x^2 + y^2 + z^2}} \vec{a}_z$ to cylindrical and spherical coordinates.
2. Define the divergence theorem and stokes theorem.
3. Two point charges with $q_1 = 2 \times 10^{-6}$ and $q_2 = -4 \times 10^{-6} \text{ C}$ are located in free space at $(1, 3, -1)$ and $(-3, 1, -2)$ respectively, in a Cartesian Coordinate system. Find the electric field $\vec{E}$ at $(3, 1, -2)$ and the force of an $8 \times 10^{-5} \text{ C}$ charge located at that point. All distances are in metres.
4. Justify the statement, "Total electrostatic field of any closed loop is zero".
5. Predict the direction of the magnetic field, when current is passing from point A to B (y direction).



6. A current loop experiences a torque in a magnetic field. Justify that the force exerted on the whole loop is zero.
7. Recall the four Maxwell equations for time-varying fields.

8. How to overcome the inconsistency of Ampere's law in the time-varying field?
9. Rearrange the Poynting vector, when the wave is propagated through a pure dielectric medium.
10. A plane wave propagating through a dielectric medium with $\epsilon_r = 8$, $\mu_r = 2$ and $E = 0.5e^{-\frac{z}{2}} \sin(10^8 t - \beta z) \hat{a}_x \text{ V/m}$. Find the phase constant and skin depth.

PART B — (5 × 13 = 65 marks)

11. (a) (i) Analyze the Gradient of scalar and divergence and curl of the vector. (6)
- (ii) Find the gradient of the scalar fields $U = x^2 y + xyz$ and $U = e^{-x} \sin 2x \cos y$. (7)

Or

- (b) (i) Convert points $P(1, 3, 5)$ and $T(0, -4, 3)$ from Cartesian to cylindrical and Spherical coordinates. (6)
 - (ii) Compute the divergence and curl of the vector field $A = yz\hat{a}_x + 4xy\hat{a}_y + y\hat{a}_z$ and evaluate it at the point $(1, -2, 3)$. (7)
12. (a) (i) If $D = (2y^2 + z)\hat{a}_x + 4xy\hat{a}_y + x\hat{a}_z \text{ C/m}^2$, find (7)
 - (1) The volume charge density is at $(-1, 0, 3)$.
 - (2) The flux through the cube is defined by $0 \leq x \leq 1, 0 \leq y \leq 1, 0 \leq z \leq 1$.
 - (3) The cube encloses the total charge.
 - (ii) Rearrange Gauss's law and develop Laplace's and Poisson's equations. (6)

Or

- (b) (i) Interpret the Electric Flux density for a uniformly charged sphere of radius ' a '. Construct a Gaussian surface for the case of $r \geq a$ and $r \leq a$ separately. (6)
- (ii) A parallel-plate capacitor has a plate area of 200 m^2 and a plate separation of 3 cm . The charge density is with air dielectric. Determine
 - (1) The capacitance of the capacitor. (4)
 - (2) The voltage between the plates (3)

13. (a) Prove that total magnetic field intensity (H) outside of the outer coaxial conductor is zero for infinitely long coaxial transmission line using Amperes law. Determine H at each Amperian path.

Or

- (b) Determine the Magnetic field and current distributions for the following three conditions
- (i) Infinite line current along the z -axis (4)
 - (ii) Infinite sheet of current (4)
 - (iii) Infinitely long coaxial transmission line (5)
14. (a) (i) A thin ring of radius 5 cm is placed on plane $z=1$ cm so that its center is at $(0, 0, 1)$ cm. If the ring carries 50 mA $\vec{a}_\phi$, find H at $(0, 0, -1)$ cm and $(0, 0, 10)$ cm. (7)
- (ii) Prove that Maxwell's equations are related to time-varying magnetic fields. (6)

Or

- (b) (i) Reconstruct Ampere's circuit law for time-varying situations to satisfy Faraday's law. (7)
- (ii) Derive the Helmholtz's wave equations for both E and H fields. (6)
15. (a) Conclude that the tangential components of $\vec{H}$ are discontinuous across the boundary, and the normal components of $\vec{H}$ are continuous across the dielectric-dielectric boundary medium. Besides, determine $\vec{H}$'s tangential and normal components across the dielectric-conductor boundary medium.

Or

- (b) (i) A uniform plane wave propagating in a lossless medium has $\vec{E} = 2 \sin[10^8 t - \beta z] \vec{a}_y$ V/m. If $\epsilon_r = 1$, $\mu_r = 2$ and $\sigma = -3$ V/m, characterize the medium. Compute the $\eta\beta$ and H . (7)
- (ii) If the wave encounters a perfectly conducting plate normal to the z -axis at $z=0$, find the reflected wave $\vec{E}_r$ and H_r . (6)

PART C — (1 × 15 = 15 marks)

16. (a) Develop the transmission and reflection coefficient expression when the incident wave from medium 2 propagates to medium 1 in normal incidence. Assume Medium 2 is air and medium 1 is Polyethylene with $\epsilon_r = 2.25$; $\mu_r = 1$. (15)

Or

- (b) Discuss the variation of flux with time in the following three ways :
- (i) A stationary loop in a time-varying magnetic field (Transformer emf) (5)
 - (ii) A time-varying loop in a static magnetic field (Motional emf) (5)
 - (iii) A time-varying loop in a time-varying magnetic field (5)

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Question Paper Code : 40984

B.E./B.Tech. DEGREE EXAMINATIONS, NOVEMBER/DECEMBER 2024.

Fourth Semester

Electronics and Communication Engineering

EC 3452 – ELECTROMAGNETIC FIELDS

(Common to : Electronics and Telecommunication Engineering)

(Regulations 2021)

Time : Three hours

Maximum : 100 marks

Answer ALL questions.

PART A — (10 × 2 = 20 marks)

1. Transform the cartesian coordinates (1, 2, 3) into spherical coordinates.
2. State Helmholtz's theorem.
3. Find the value of k , if the electric flux density in a charge free region is given by $D = 5x\bar{a}_x + 10y\bar{a}_y - kz^2\bar{a}_z$.
4. Determine the energy stored in parallel plate capacitor of 10cm by 10cm has a separation of 1 cm and a voltage difference of 10V.
5. State Biot Savart's law.
6. Distinguish between magnetic scalar potential and magnetic vector potential.
7. In a material for which $\sigma = 5 \text{ s/m}$, $\epsilon_r = 1$ and $E = 100 \sin 10^6 t \text{ v/m}$. Find its displacement current density.
8. State Guass's law for magnetic field and express in integral form.
9. Define group velocity. Write its expression.
10. Calculate the characteristic impedance of free space.

PART B — (5 × 13 = 65 marks)

11. (a) State and verify Divergence theorem for its vector $A = 4x\bar{a}_x - 2y^2\bar{a}_y + z^2\bar{a}_z$ taken over the cube bounded by $x=0$, $x=1$, $y=0$ and $y=1$.

Or

- (b) State and verify Stoke's theorem. Give its application.

12. (a) Derive

(i) Laplace equation (7)

(ii) Equation of continuity. (6)

Or

- (b) Calculate the capacitance of a parallel plate capacitor with two dielectrics ($\epsilon_{r1} = 2$ of thickness 1 mm, $\epsilon_{r2} = 1$ of thickness 2 mm) kept parallel to the plates of area of 50 cm². What is the voltage gradient in each dielectric if 100 V is applied across its plates.

13. (a) Derive an expression for the magnetic field intensity on the axis of solenoid.

Or

- (b) Derive expression for inductance of

(i) solenoid (7)

(ii) toroid (6)

14. (a) State and derive Maxwell's equation in integral form.

Or

- (b) Derive wave equation for conducting medium.

15. (a) Obtain the expression for reflection coefficient and transmission coefficient when a wave incident normally at a dielectric boundary.

Or

- (b) State and prove Poynting theorem.

PART C — (1 × 15 = 15 marks)

16. (a) Show that the displacement current through the parallel plate capacitor of capacitance C is equal to the conduction current I if a supply voltage $V = V_m \sin \omega t$ is applied across the capacitor.

Or

- (b) A uniform plane wave is travelling at a velocity of 2.5×10^8 m/s having a wavelength of 0.5 mm in a non-magnetic good conductor. Calculate the frequency of wave and conductivity of the medium.
